TEXORA · LECTURE SHEET

Activity Sampling

Estimating how time is really spent — instantaneous random observations that turn proportions of working, idle and delay states into statistically defensible figures, with the pilot study, the sample-size formula and the interpretation of results.

TE-4107 · PRODUCTION PLANNING AND CONTROL · TOPIC 4 · LEVEL 4 TERM 1

1. Definition and principle

Activity sampling (work sampling, ratio-delay study) is a work-measurement technique in which a large number of instantaneous observations are made at random intervals, recording what each worker or machine is doing *at that instant* — working, idle, waiting for material, under repair, and so on. The underlying statistical principle: the proportion of observations falling in a state estimates the proportion of total time spent in that state, with an accuracy that improves with the number of observations. If looms are seen stopped for warp breaks in 18 of 100 random glances, they spend about 18 % of time so stopped.

The technique suits what stopwatch study cannot touch: long-cycle work, irregular work, group activities, and utilisation/delay questions spread over whole departments and whole weeks. It is also less intrusive — no observer stands with a watch — so behaviour distortion is smaller, though the first days of any study still show a "being watched" effect that the pilot study absorbs.

2. Procedure

1. **Define the objective and the activity categories** precisely — e.g. for looms: running; stopped–warp break; stopped–weft break; stopped–waiting for beam/material; stopped–mechanical/electrical repair; stopped–no operator. Vague categories produce vague data.
2. **Pilot study**: a small preliminary run (a few hundred observations) to estimate the key proportion p, test the categories and the route, and train observers.
3. **Calculate the required number of observations** for the desired accuracy (Section 3).
4. **Fix random observation times** (random number tables or generated random times) across the whole study period and all shifts — never a comfortable fixed round, which synchronises with cyclic events and biases the result.
5. **Make the tours**: at each instant, record the state of every subject on the route at first sight — the state when first seen, not after a moment's watching.

6. **Analyse and interpret**: compute proportions with their confidence limits, convert to hours and costs, and compare with standards (Section 4).



Bar widths proportional to observed shares — the running share is the shed's utilisation picture at a glance.

Fig. 1 — Typical activity-sampling result for a loom shed: the non-running third, not the running two-thirds, is where improvement money hides.

3. Sample size and accuracy

For a proportion p (in %), the standard error of the estimate after N observations is $\sigma = \sqrt{(p\,(100 - p) \div N)}$. At 95 % confidence the result is quoted as $p \pm 2\sigma$. Setting the acceptable error L (in %) gives the required sample size:

$$N = 4 \times p \times (100 - p) \div L^2$$

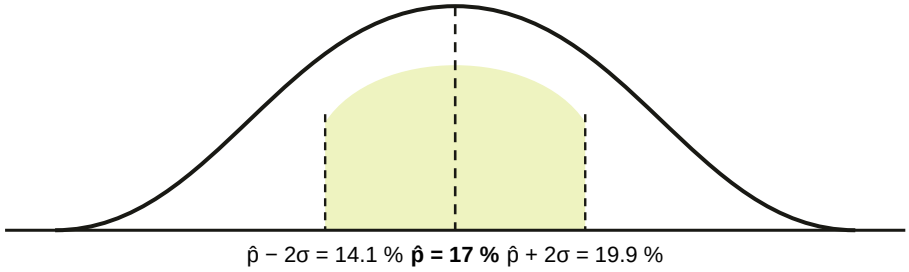
WORKED EXAMPLE — HOW MANY OBSERVATIONS?

Pilot study: 200 observations of looms; 36 found stopped for all causes other than running normally on the studied state "idle/waiting and stopped" gives $p = 36 \div 200 \times 100 =$ **18 %**.
Required accuracy: ±3 % at 95 % confidence → $L = 3$.
$N = 4 \times 18 \times 82 \div 3^2 = 5{,}904 \div 9 =$ **656 observations**.
At 40 looms per tour, $656 \div 40 \approx 17$ tours, spread randomly over days and shifts.
Check after the study: final result p = 17 %. $\sigma = \sqrt{(17 \times 83 \div 656)} = \sqrt{2.151} \approx 1.47$ % → the result is reported as **17 % ± 2.9 %** (95 % confidence) — inside the ±3 % specified. If it were not, more observations would be added: accuracy is bought with N, and only with N.

Note the shape of the formula: proportions near 50 % need the largest samples; rare events (p = 2 %) need modest N for the same *absolute* error but huge N for the same *relative* error. The planner must state which accuracy matters before the study, not after.

4. Interpretation of results

Results are interpreted in money and time, not in percentages alone. If sampling shows a 60-loom shed's looms wait for weft/material 5 % of a 16-hour day, that is 0.05 × 16 × 60 = 48 loom-hours lost daily — the size of three extra looms, recoverable by material logistics rather than by buying looms. Interpretation rules:



Sampling distribution of p̂: about 95 % of repeated studies would land inside the shaded ±2σ band.

Fig. 2 — The confidence band of the worked example: the estimate p̂ = 17 % with its 95 % limits, 17 % ± 2.9 %.

- Compare proportions with the confidence interval attached — a change from 17 % to 15 % between two studies whose intervals are ±3 % may be no change at all.
- Break results down by shift, day and section before averaging them away: a delay concentrated on night shift is a supervision/materials story, not a machine story.
- Distinguish **avoidable** from **unavoidable** delay: unavoidable delay feeds allowances in standard times; avoidable delay feeds method study and management action.
- Sampling measures *proportions of time*, not performance rate: a worker "working" in every observation may still work slowly. Sampling never sets an elemental standard time; it sizes the states within which other techniques work.

Glossary

Activity sampling — estimation of time proportions from random instantaneous observations. **Pilot study** — preliminary small study to estimate p, test categories and train observers. **Proportion p** — percentage of observations (hence of time) in a state. **Confidence limit** — the ± range within which the true proportion lies with stated probability (95 % ⇒ ±2σ). **Instantaneous observation** — recording the state at first sight at a random instant. **Avoidable delay** — delay removable by management action, as opposed to unavoidable delay allowed for in standards.

Key points

• Observation proportions estimate time proportions; accuracy grows only with the number of observations.

• $N = 4p(100 - p)/L^2$: pilot p = 18 %, L = 3 → N = 656; final 17 % ± 2.9 % at 95 % confidence.
• Observation times must be random across shifts; fixed rounds bias results.
• Interpret in hours and money, split by shift/section, and separate avoidable from unavoidable delay.
• Sampling sizes states; it does not rate pace or set elemental standard times.

Practice

1. A pilot study of 250 observations finds operators waiting for work in 50. How many observations are needed for ±2 % accuracy at 95 % confidence on this proportion?
2. After 900 observations a state shows p = 10 %. Compute σ and the 95 % confidence statement.
3. Why must observation instants be random rather than hourly on the hour?
4. A sampling study reports sewing operators "absent from workplace" 6 % of the time (±1.5 %). The manager proposes disciplining the section. Criticise the proposal on interpretation grounds.

Answer key

1. p = 50 ÷ 250 × 100 = 20 %. N = 4 × 20 × 80 ÷ 2^2 = 6,400 ÷ 4 = 1,600 observations.
2. σ = √(10 × 90 ÷ 900) = √1 = 1 %. Result: 10 % ± 2 % at 95 % confidence.
3. Work is cyclic — breaks, material deliveries, shift events and machine cycles repeat on clock patterns. Fixed hourly observations synchronise with those cycles (always catching, or always missing, the same phase) and bias the proportions. Random instants sample all phases fairly.
4. "Absent" is not a diagnosis: the category includes toilet and water breaks (legitimate personal time covered by allowances), fetching materials (a methods/logistics fault), and waiting at a shared facility — only a fraction is unauthorised absence. The correct response is to break the category down in a follow-up study and fix causes (materials supply, facilities) before any question of discipline; a 6 % figure with ±1.5 % also means the true value may be 4.5 %, much of which allowances already recognise.